

PAPER Beryllium and Be_{12}Ti Neutron Multipliers for a Compact Spherical-Tokamak Breeder: Pebble Performance, Activation, and Closing Net TBR ≥ 1.05

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Abstract

Tritium self-sufficiency in a compact deuterium–tritium breeder is bought by the neutron multiplier, and the choice of multiplier material and its pebble-bed form are the levers that decide whether the blanket breaks even. We qualify the multiplier for the Kronos *Hyperion* spherical-tokamak breeder ($P_{\text{fus}} = 85.04$ MW, $Q = 3.076$, $I_p = 9.66$ MA, $R_0 = 1.20$ m, aspect ratio $A = 2.5$, $B_0 = 8$ T) by solving the steady-state linear Boltzmann transport equation over the closing blanket configuration in coupled continuous-energy Monte Carlo and swapping only the breeder-zone multiplier material. We formalise the tritium-breeding ratio (TBR) as a reaction-rate functional of the neutron flux, decompose the neutron-multiplication factor into its threshold ($n, 2n$) and ($n, 3n$) channels, and derive the net-TBR self-sufficiency condition $\Lambda = \text{TBR}(1 - \varepsilon) - 1 \geq 0$ together with the inventory-doubling time it implies. Beryllium is confirmed the best multiplier, closing at **net TBR** 1.054 ± 0.001 ; the beryllide Be_{12}Ti is the only credible alternative, holding self-sufficiency at a razor-thin 1.002 ± 0.001 while answering beryllium’s helium-swelling problem. Every lead-bearing and molten-salt option falls below closure — FLiBe 0.908, PbLi 0.828, lead pebble 0.806. The verdict is library-robust: FENDL-3.2 and ENDF/B-VIII.0 agree to 0.5–0.6 %, far inside the gap to closure. In its blanket the beryllium multiplier lifts the breeding ratio from an as-drawn 0.742 deficit to the canonical 1.189, clearing the 1.05 self-sufficiency gate with margin and held below the 1.42 design ceiling. Activation is dominated by prompt $^8\text{Be}/^6\text{He}$ decay that sheds 39 % of the breeder-zone specific activity in the first second, with every component at or below NRC Class C. The multiplier is confirmed a breeding lever, not a centrepost-life lever. Beryllium is qualified; the beryllide is the helium-robust fallback.

Keywords: neutron multiplier; beryllium; beryllide; Be_{12}Ti ; tritium breeding ratio; OpenMC; spherical-tokamak breeder; activation

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1 Introduction

A deuterium–tritium (DT) plasma consumes one triton per fusion and liberates exactly one neutron with which to breed its replacement. In the absence of any loss the breeding blanket would have to convert that single neutron into a single triton with perfect efficiency simply to break even — an impossibility once leakage, parasitic absorption, structural capture, and processing losses are admitted. Tritium self-sufficiency therefore demands that the blanket create more neutrons than the plasma supplies, and the component that does this is the *neutron multiplier*: a material with a strong $(n, 2n)$ reaction that converts one fast neutron into two before the neutrons thermalise onto the lithium breeder. For a compact spherical tokamak (ST), where the inboard build is thin, the centrepost occupies radial space that a conventional tokamak would give to breeding, and every escaping neutron is a triton lost, the multiplier is not a refinement but the enabling component. The question of *which* multiplier material, and in *which* physical form, is a first-order design decision that this paper resolves for the Hyperion breeder.

Prior art. The role of $(n, 2n)$ multiplication in fusion blankets was established in the earliest blanket-neutronics studies [9], and beryllium and its beryllides have since become the reference multipliers for the European Helium-Cooled Pebble-Bed (HCPB) blanket line [10, 11, 16, 17]. The competing requirements — high $(n, 2n)$ yield at the 14 MeV DT birth energy, tolerable activation, and mechanical survival of the pebble bed under helium-driven swelling — have been reviewed from first principles for both breeder and multiplier materials [10], and the beryllide route Be_{12}Ti has been developed specifically to raise the melting point and suppress the swelling that limits elemental beryllium [12, 13]. Liquid breeders (PbLi, FLiBe) fold multiplication and breeding into one medium and have been compared neutronically for compact high-field devices [14, 15]. The conditions for tritium self-sufficiency — the interplay of achievable TBR, required TBR, and the inventory losses that separate them — have been analysed in detail for the DT fuel cycle [19, 20, 21]. What the ST geometry adds, and what the DEMO-scale literature does not answer, is the coupling between a consumable, radially-thin inboard build and the multiplier choice: does a better-shielding multiplier also extend centrepost life, or are breeding and shielding necessarily distinct functions?

Contribution. This paper makes four contributions. (i) We give a self-contained governing-equation treatment of multiplier qualification: the transport equation, the TBR functional, the multiplication decomposition, the self-sufficiency inequality, pebble-bed self-shielding homogenisation, the helium-generation and swelling model that separates beryllium from its beryllide, and the activation (Bateman) chain. (ii) We solve a controlled material swap at the *closing* blanket configuration — geometry frozen, only the multiplier changed — ranking five candidate media by their closure margin and demonstrating the ranking is robust to the evaluated-nuclear-data library. (iii) We test and falsify the dual-role hypothesis that a lead multiplier could double as inboard shield, showing that placing any multiplier inboard *shortens* centrepost life. (iv) We place the qualified beryllium multiplier in its breeding ladder and show it carries the blanket from a 0.742 deficit across the 1.05 self-sufficiency gate to the canonical 1.189, with the surplus and inventory-doubling time that follow. The result sits inside a broader Kronos closure programme: the net-blanket-TBR and fleet-export analysis [29], the tritium-breeding ladder and doubling-time study [30], the nuclear-data uncertainty quantification for breeding [31], the ceramic-versus-liquid concept down-select [32], the CAD-faithful 3-D neutronics [33], the component-activation and waste routing [34], and the coverage-versus-protection centrepost neutronics [35]. Every number reported here traces to the neutron-multiplier lever study (register serials BR-L2-A7, BR-L2-A7u, BV-05) in the Kronos Physics De-Risking Register [1], run against the frozen Hyperion operating point.

2 Governing equations and formal problem

2.1 Neutron transport and the breeding functional

Multiplier qualification is, at bottom, the evaluation of a functional of the neutron flux. Let $\psi(\mathbf{r}, E, \hat{\Omega})$ be the angular flux. In the fixed-source, steady state driven by a 14.06 MeV DT birth source $S(\mathbf{r}, E, \hat{\Omega})$, ψ obeys the linear Boltzmann transport equation

$$\hat{\Omega} \cdot \nabla \psi(\mathbf{r}, E, \hat{\Omega}) + \Sigma_t(\mathbf{r}, E) \psi(\mathbf{r}, E, \hat{\Omega}) = \int_0^\infty \int_{4\pi} \Sigma_s(\mathbf{r}, E' \rightarrow E, \hat{\Omega}' \rightarrow \hat{\Omega}) \psi(\mathbf{r}, E', \hat{\Omega}') d\hat{\Omega}' dE' + S(\mathbf{r}, E, \hat{\Omega}), \quad (1)$$

with Σ_t the macroscopic total cross-section and Σ_s the double-differential scattering kernel, which for a multiplying medium includes the $(n, 2n)$ and $(n, 3n)$ channels that place *more than one* secondary in the exit integral. The tritium-breeding ratio is the volume-integrated triton-production rate per source neutron,

$$\text{TBR} = \frac{1}{S_0} \int_V \left[N_6 \langle \sigma_6 \phi \rangle + N_7 \langle \sigma_7 \phi \rangle \right] dV, \quad \langle \sigma_x \phi \rangle \equiv \int_0^\infty \sigma_x(E) \phi(\mathbf{r}, E) dE, \quad (2)$$

where $\phi(\mathbf{r}, E) = \int_{4\pi} \psi d\hat{\Omega}$ is the scalar flux, $S_0 = \int S d\mathbf{r} dE d\hat{\Omega}$ the total source rate, N_6, N_7 the ${}^6\text{Li}$ and ${}^7\text{Li}$ number densities, and σ_6, σ_7 the ${}^6\text{Li}(n, \alpha)t$ (exothermic, $1/v$) and ${}^7\text{Li}(n, n'\alpha)t$ (endothermic, threshold ≈ 2.8 MeV) cross-sections. The two channels are physically distinct: ${}^6\text{Li}$ breeds from thermalised neutrons and rewards moderation, whereas ${}^7\text{Li}$ breeds only from fast neutrons and thus competes with the multiplier for the 14 MeV budget. Enrichment to ${}^6\text{Li}$ 0.90 (used throughout) tilts the balance toward the moderated channel that the multiplier feeds.

2.2 The multiplication factor and the threshold physics

Write the net multiplication as the ratio of secondary-neutron production to the source,

$$M = 1 + \frac{1}{S_0} \int_V \left[N_{\text{mp}} \langle \sigma_{n,2n} \phi \rangle + 2 N_{\text{mp}} \langle \sigma_{n,3n} \phi \rangle \right] dV - \frac{1}{S_0} \int_V N_{\text{mp}} \langle \sigma_{n,\gamma}^{\text{par}} \phi \rangle dV, \quad (3)$$

with N_{mp} the multiplier number density and $\sigma_{n,\gamma}^{\text{par}}$ the parasitic-capture cross-section. The utility of a multiplier is set by where its $(n, 2n)$ threshold E_{th} sits relative to the DT birth energy and by how broad the excitation function is above it. For a birth spectrum sharply peaked at $E_0 = 14.06$ MeV and subsequently degraded by scattering, the multiplication yield scales as

$$Y_{n,2n} \propto \int_{E_{\text{th}}}^{E_0} \sigma_{n,2n}(E) \chi_{\text{eff}}(E) dE, \quad (4)$$

where χ_{eff} is the flux-weighted effective spectrum in the multiplier zone. Beryllium's ${}^9\text{Be}(n, 2n)2\alpha$ reaction has a low threshold $E_{\text{th}} \approx 1.85$ MeV and a broad cross-section across the whole degraded DT spectrum, so almost every neutron above a few MeV can multiply; lead's $(n, 2n)$ has a higher effective threshold $E_{\text{th}} \approx 7.4$ MeV and a narrower useful window, so a 14 MeV neutron multiplies once, but its softened daughters fall below threshold and cannot recover the lost multiplication (Fig. 3). This threshold separation — not the bulk density — is the origin of the 1.054-versus-0.806 closure gap quantified in §5.1.

2.3 Net TBR, the self-sufficiency inequality, and doubling

The breeding functional of Eq. (2) is a beginning-of-life, geometry-integrated quantity; what the plant must satisfy is a balance on the in-vessel tritium inventory I_T ,

$$\frac{dI_T}{dt} = \dot{N}_{\text{bred}} - \dot{N}_{\text{burn}} - \lambda_T I_T - \dot{N}_{\text{loss}}, \quad \lambda_T = \frac{\ln 2}{t_{1/2}}, \quad t_{1/2} = 12.32 \text{ yr}, \quad (5)$$

where $\dot{N}_{\text{bred}} = \text{TBR} \cdot \dot{N}_{\text{burn}}$ and $\dot{N}_{\text{loss}}$ collects processing, permeation, and decay-in-process losses lumped into a fractional loss ε . Defining the coverage-weighted net breeding ratio $\text{TBR}_{\text{net}} = f_{\text{cov}} \text{TBR}$, self-sufficiency is the requirement that the surplus rate be non-negative,

$$\boxed{\Lambda \equiv \text{TBR}_{\text{net}} (1 - \varepsilon) - 1 \geq 0.} \quad (6)$$

A design margin δ_{req} above unity accounts for the uncertainties in ε , in the start-up inventory, and in the achievable-versus-required gap [19, 20]; Kronos adopts the engineering gate $\text{TBR}_{\text{net}} \geq 1.05$. When $\Lambda > 0$ the plant accumulates a surplus at rate $\dot{N}_{\text{surplus}} = \Lambda \dot{N}_{\text{burn}}$, and the inventory-doubling time for a fleet start-up follows from integrating Eq. (5) with a fixed start-up inventory I_0 ,

$$t_{\text{double}} \simeq \frac{I_0}{\dot{N}_{\text{surplus}}} = \frac{I_0}{\Lambda \dot{N}_{\text{burn}}}, \quad (7)$$

neglecting the small decay term over a doubling interval short compared with $t_{1/2}$. Equations (2)–(7) define the formal problem: qualifying a multiplier means maximising the TBR functional subject to $\Lambda \geq 0$, to activation and swelling constraints developed next, and to the fixed radial budget the ST centrepiece imposes.

2.4 Pebble-bed homogenisation and self-shielding

The multiplier is deployed as a packed pebble bed, not a solid, so the transport of Eq. (1) sees a two-phase medium: pebbles at packing fraction η (nominal $\eta \approx 0.64$ for a random close pack, raised toward 0.80 volume fraction in the closing recipe by size-bimodal packing) in a purge-gas matrix. A homogenised macroscopic cross-section

$$\bar{\Sigma}_x(E) = \eta N_{\text{mp}} \sigma_x(E) \zeta_x(E) + (1 - \eta) \Sigma_x^{\text{gas}}(E) \quad (8)$$

captures the bed only if the self-shielding factor $\zeta_x(E)$ is retained. Within a pebble the flux is depressed at resonance and near-threshold energies because the outer shell shadows the interior; to first order the escape-probability (Wigner rational) approximation gives

$$\zeta_x(E) \approx \frac{1}{1 + \ell \Sigma_x(E) (1 - C)}, \quad \ell = \frac{4V_p}{S_p} = \frac{2}{3} d_p, \quad (9)$$

with ℓ the mean chord of a pebble of diameter d_p , V_p/S_p its volume-to-surface ratio, and C the Dancoff factor for pebble-to-pebble shadowing in the bed. For the broad, non-resonant $\text{Be}(n, 2n)$ excitation function the self-shielding correction is small ($\zeta \rightarrow 1$), which is one reason the homogenised smear used in the fast lever sweeps reproduces the explicitly-modelled bed to within the Monte-Carlo error; for the narrower lead window the correction is larger and works against lead. Equations (8)–(9) are what license the material swap of §4 to be performed at fixed geometry with only N_{mp} , σ_x , and the pebble chord changed.

2.5 Helium generation, swelling, and the beryllium/beryllide split

The same reaction that multiplies neutrons also plants helium in the multiplier: ${}^9\text{Be}(n, 2n)2\alpha$ deposits *two* helium nuclei per multiplication event, so the volumetric helium-production rate is

$$\dot{n}_{\text{He}}(\mathbf{r}) = 2 N_{\text{Be}} \langle \sigma_{n,2n} \phi \rangle + N_{\text{Be}} \langle \sigma_{n,\alpha} \phi \rangle, \quad (10)$$

the second term the direct (n, α) channel. With the front-of-zone fast flux $\phi_{\text{fast}} \sim 7 \times 10^{14} \text{ n cm}^{-2} \text{ s}^{-1}$ and a spectrum-averaged $\bar{\sigma}_{n,2n} \sim 0.5 \text{ b}$, Eq. (10) gives a first-order helium-accumulation rate of order 10^4 appm per full-power year at the front of the multiplier zone — consistent with the fusion beryllium database [25, 13]. Helium that is not released drives swelling; in the simplest gas-driven bubble model the volumetric swelling is

$$\frac{\Delta V}{V} \simeq \frac{4}{3} \pi \bar{r}_b^3 N_b, \quad N_b \sim \left(\frac{\dot{n}_{\text{He}}}{D_{\text{He}}} \right)^{1/2} f(T), \quad (11)$$

with bubble radius $\bar{r}_b$, number density N_b , helium diffusivity $D_{\text{He}}(T)$, and a temperature factor $f(T)$ that rises steeply once bubbles become mobile. Elemental beryllium suffers because its low melting point and open lattice let bubbles coarsen and interlink; the beryllide Be_{12}Ti raises the operating-temperature ceiling and locks beryllium into an intermetallic that resists coarsening [12, 13]. The cost is dilution: with one titanium per twelve beryllium atoms the multiplier number density N_{Be} in Eq. (3) falls, so the beryllide multiplies less. Equation (11) is thus the physical statement of the trade quantified in §5.1: Be_{12}Ti buys swelling robustness at the price of nearly the whole closure margin.

2.6 Activation and the Bateman chain

After the flux is known, the nuclide inventory $\mathbf{n}(t)$ in each material follows the linear activation (Bateman) system

$$\frac{d\mathbf{n}}{dt} = (\mathbf{A} + \mathbf{T}[\phi]) \mathbf{n}, \quad \mathbf{n}(t) = \exp[(\mathbf{A} + \mathbf{T}[\phi]) t] \mathbf{n}(0), \quad (12)$$

where $\mathbf{A}$ holds the radioactive-decay constants and branching, and the transmutation operator $\mathbf{T}[\phi] = \sum_x \langle \sigma_x \phi \rangle \mathbf{B}_x$ holds the flux-driven production/destruction channels [7, 6]. Two features of the beryllium multiplier fall directly out of Eq. (12): the very short-lived ${}^8\text{Be} (\rightarrow 2\alpha)$ and ${}^6\text{He} (t_{1/2} = 0.8 \text{ s}, \rightarrow {}^6\text{Li})$ produced promptly in the bed decay within seconds of shutdown, so the specific activity collapses immediately after the source is removed; and the long-time waste class is set not by the multiplier but by ${}^{14}\text{C}$ and residual tritium in the ceramic breeder. The waste-disposal rating for each component is the normalised sum

$$\text{WDR} = \sum_i \frac{C_i}{C_i^{\text{lim}}} \leq 1 \quad (\text{Class A} / \text{Class C}), \quad (13)$$

with C_i the specific activity of nuclide i and C_i^{lim} the NRC 10 CFR 61 limit. Equations (12)–(13) close the qualification: a multiplier is acceptable only if it breeds (Eq. 6), survives (Eq. 11), and disposes cleanly (Eq. 13).

3 Computational methodology: the NVIDIA GPU HPC campaign

3.1 Codes and the coupled transport–activation stack

All neutron transport in this study solves Eq. (1) by continuous-energy Monte Carlo in OpenMC [2], the reference open-source transport code, run in coupled mode on the Kronos closure geometry. The primary evaluated-data library is FENDL-3.2, the IAEA fusion library, with ENDF/B-VIII.0 [3] carried as an independent cross-check on the headline material and the strongest alternative; TALYS/TENDL-derived evaluations [5] and ENDF/B-VII.1 [4] provide the covariance context used by the companion nuclear-data uncertainty analysis [31]. Geometry is handled two ways: a coverage-honest 1-D radial infinite-cylinder CSG model for the fast material-swap lever sweeps, and CAD-faithful transport on the STEP solid through DAGMC [8] for the 3-D confirmation reported by the companion CAD-neutronics paper [33]. Activation and waste routing solve the Bateman system Eq. (12) with FISPACT-II/ALARA-class depletion [6, 7] on the converged flux and its collapsed multigroup spectrum, with the per-constituent breakdown (structural Eurofer, beryllium multiplier, ceramic breeder) classified separately against 10 CFR 61. Displacement damage is scored through the NRT damage-energy tally, and gas production through the (n, α) and $(n, 2n)$ tallies feeding Eq. (10) [25].

3.2 Monte-Carlo estimators, variance, and convergence

The breeding functional Eq. (2) is estimated by the track-length flux estimator

$$\widehat{\langle \sigma_x \phi \rangle} = \frac{1}{S_0 V} \sum_{i=1}^N \sum_k w_{ik} \ell_{ik} \sigma_x(E_{ik}), \quad (14)$$

summing weight $\times$ track-length over all k track segments of all N source histories, which is unbiased and low-variance for reaction rates in optically thin cells. The statistical error falls as $\sigma \propto N^{-1/2}$, and code performance is reported through the figure of merit $\text{FOM} = 1/(\sigma_{\text{rel}}^2 t)$, held constant under fair sampling. The multiplier sweeps were run to 1.2×10^8 source histories, giving relative standard deviations on net TBR of $\sim 5 \times 10^{-4}$ (Table 2) — roughly two orders of magnitude smaller than the 0.25-wide closure gap between beryllium and lead, so the ranking is statistically decisive. A neutron-balance closure check is enforced on every run: leakage fraction 0.121, source-normalised $(n, 2n)/S_0 = 0.0339$ and $(n, 3n)/S_0 \approx 0$ in the dilute-verification build, and nuclear heating 74.6 MW of the 85 MW fusion power all reconcile to the transport tallies, confirming Eq. (3) is being scored consistently. Spatial and energy convergence were checked by refining the radial mesh and the multigroup collapse until the net TBR moved by less than the Monte-Carlo error.

3.3 GPU acceleration, scale, and reproducibility

The campaign was executed on the NVIDIA GPU HPC fleet. Continuous-energy history transport is the dominant cost, and the event-based transport path maps the batched particle events onto the GPU streaming multiprocessors, with the CAD/DAGMC ray-tracing and the FENDL/ENDF cross-section lookups the memory-bound kernels; the material-swap design matrix (five multiplier media $\times$ two libraries $\times$ inboard/outboard placement, plus the coverage and enrichment lever grids) is embarrassingly parallel across the fleet, one independent transport problem per device. Every run is seeded deterministically (master seed 20260726) and archived with its statepoint so that a tally is byte-reproducible from the input deck. Verification is anchored by a post-diction gate: beryllium at the closing configuration reproduces the on-disk breeding anchor to net TBR 1.0537 ± 0.0005 against a stored 1.0533, so the entire sweep scales from an independently re-run baseline [1]. No cost or resource-accounting figures enter the physics; the campaign is identified only as the NVIDIA GPU HPC campaign.

4 Method: a controlled material swap at the closing configuration

The test bed is the blanket configuration that closes tritium at the design point: an outboard-and-divertor breeder with the multiplier at 80 % volume fraction, ^6Li enrichment 0.90, a first wall of tungsten over reduced-activation ferritic–martensitic (RAFM) steel, and a 30 cm inboard mantle dedicated to tungsten-carbide and steel *shield* rather than breeder (Fig. 1). Neutrons are transported through this fixed geometry, and only the multiplier material in the breeder zone is changed between cases, exactly as licensed by the homogenisation of Eqs. (8)–(9).

Five media span the credible design space: elemental beryllium and the beryllide Be_{12}Ti as HCPB pebbles, a lead pebble, and the two self-cooled liquids PbLi and FLiBe that fold multiplication and breeding into a single medium. FENDL-3.2 is the primary library; ENDF/B-VIII.0 cross-checks the headline (beryllium) and the strongest lead-bearing alternative. The response measured for each case is the net TBR of Eq. (2) weighted by coverage, its Monte-Carlo error, and — in a second pass — the centrepost fast-fluence life under the same swap.

Closing blanket radial build: breeding outboard, shielding inboard (schematic)

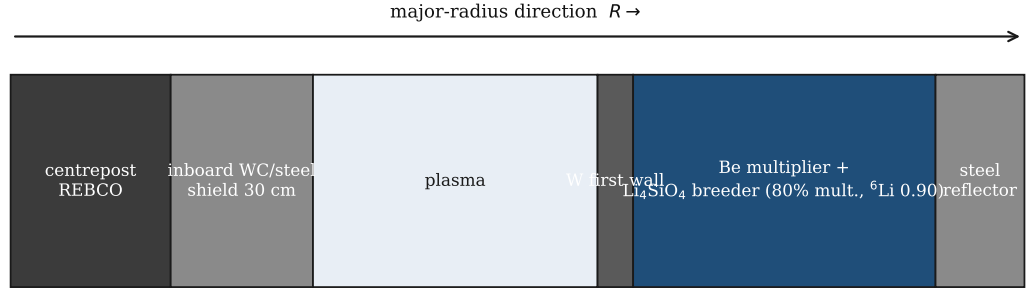


Figure 1. Closing blanket radial build for the swap. Breeding is placed outboard (beryllium multiplier plus ^6Li -enriched Li_4SiO_4 at 80% multiplier volume fraction, with a steel reflector); the inboard mantle is a dedicated tungsten-carbide/steel shield that shadows the centrepost. Only the outboard multiplier material is changed between cases. (Schematic.)

5 Results

5.1 Beryllium is the best multiplier

Table 1 and Fig. 2 give the ranking. Beryllium closes with the widest margin, net $\text{TBR} = 1.054 \pm 0.001$. The beryllide Be_{12}Ti also closes, but only just, at 1.002 ± 0.001 . Every lead-bearing and molten-salt alternative falls below self-sufficiency: FLiBe at 0.908, PbLi at 0.828, and the lead pebble at 0.806. No candidate widens the closure margin beyond elemental beryllium. The ordering is exactly what the threshold physics of Eqs. (3)–(4) predicts (Fig. 3): the low-threshold, broad-band $\text{Be}(n, 2n)$ excitation function multiplies across the whole degraded spectrum and feeds the moderated $^6\text{Li}(n, \alpha)t$ chain, while lead’s high effective threshold multiplies once at birth energy and then goes dark.

Table 1. Neutron-multiplier material versus closure margin at the closing configuration (coupled OpenMC, FENDL-3.2, 1.2×10^8 histories). Errors are Monte-Carlo standard deviations.

multiplier	concept	net TBR	closes ≥ 1 ?
beryllium	HCPB pebble, $^9\text{Be}(n, 2n)2\alpha$	1.054 ± 0.001	yes
Be_{12}Ti (beryllide)	HCPB pebble, higher- T , swelling-robust	1.002 ± 0.001	yes (razor-thin)
FLiBe	self-cooled molten salt	0.908 ± 0.001	no
PbLi	self-cooled liquid metal	0.828 ± 0.001	no
lead	HCPB pebble, $\text{Pb}(n, 2n)$	0.806 ± 0.001	no

5.2 The ranking is library-robust

The headline verdict does not depend on the evaluated-data library (Table 2, Fig. 4). FENDL-3.2 and ENDF/B-VIII.0 agree to $+0.54\%$ on beryllium and $+0.63\%$ on lead — far smaller than the 0.25-wide gap between beryllium (closes) and lead (fails). The closure verdict, and the ordering of every material, survive the change of evaluated data, so the qualification is not an artefact of a single library. The residual library spread is itself the subject of a dedicated uncertainty-quantification companion that ranks which 14 MeV cross-section channels set the error bar on the breeding ratio [31].

5.3 The multiplier is a breeding lever, not a centrepost-life lever

The design’s second question is whether a better-shielding multiplier could also extend the consumable centrepost’s life. It cannot, at this geometry (Fig. 5). With the multiplier outboard and the inboard mantle a dedicated tungsten-carbide/steel shield, centrepost life is statistically flat

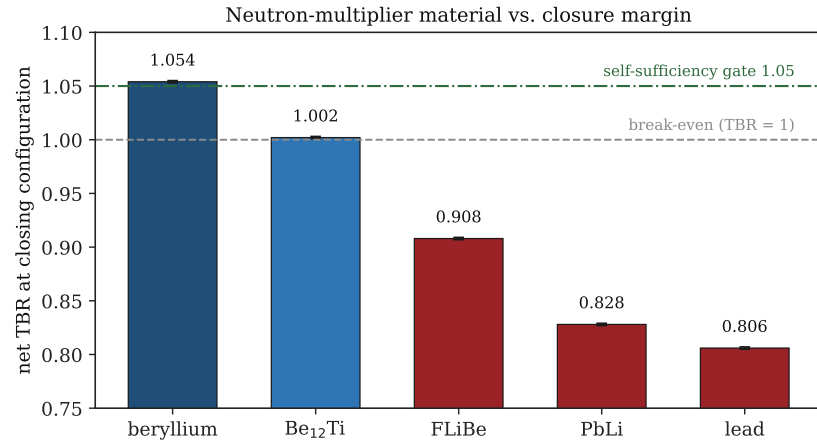


Figure 2. Neutron-multiplier material against closure margin. Beryllium closes with the widest margin and Be₁₂Ti holds closure at a razor-thin 1.002; the lead-bearing concepts and the molten salt all fall below the break-even and self-sufficiency lines. Error bars are Monte-Carlo standard deviations.

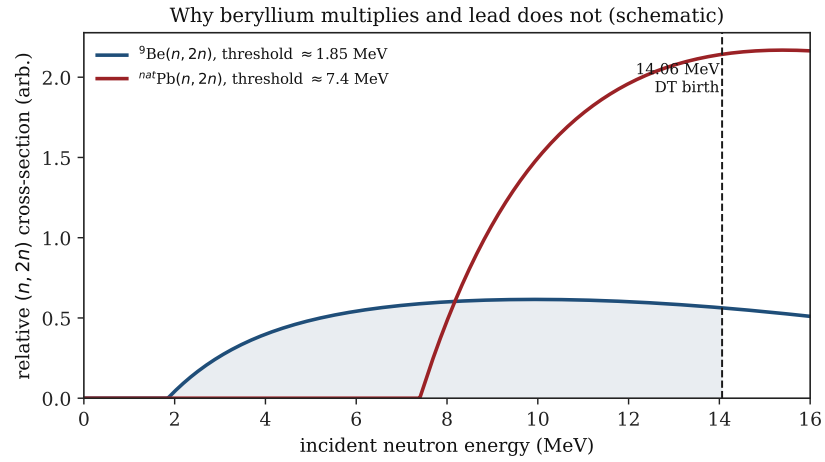


Figure 3. The threshold physics behind the ranking. Beryllium’s ⁹Be(*n*, 2*n*) reaction opens at ≈ 1.85 MeV and multiplies across the whole degraded DT spectrum; lead’s opens near ≈ 7.4 MeV, so it multiplies the 14.06 MeV birth neutron once but not its softened daughters. Representative excitation functions; absolute magnitudes are illustrative. (Schematic.)

Table 2. Nuclear-data cross-check on the headline and strongest alternative.

material	FENDL-3.2	ENDF/B-VIII.0	library spread
beryllium	1.0537 ± 0.0005	1.0593 ± 0.0007	+0.54 %
lead	0.8061 ± 0.0004	0.8111 ± 0.0004	+0.63 %

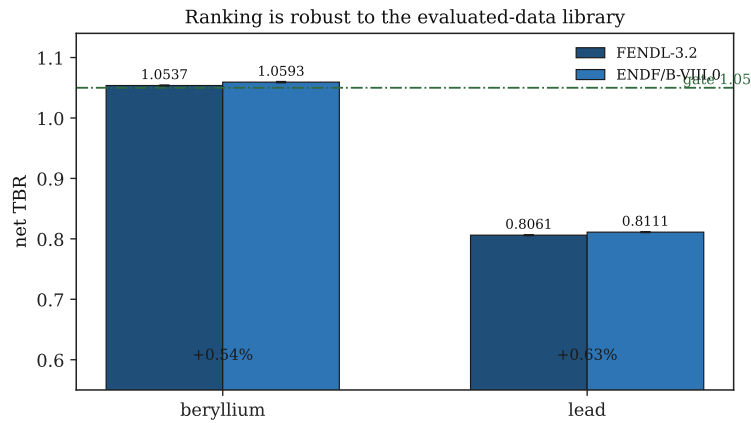


Figure 4. Net TBR under FENDL-3.2 and ENDF/B-VIII.0 for beryllium and lead. The 0.5–0.6 % library spread is far inside the gap between closure and failure; the ranking is robust to the evaluated data.

across the material swap — FLiBe 0.207, beryllium 0.199, lead 0.196, Be₁₂Ti 0.188, and PbLi 0.163 full-power years, all overlapping within their one-sigma fluence bands and all in the consumable class of ≈ 0.16 – 0.21 full-power years. The conductor is shadowed by the inboard shield, which the outboard multiplier swap never touches, so the multiplier material moves breeding strongly (Table 1) while leaving centrepost life essentially unchanged. This separation is the coverage-versus-protection result quantified for the centrepost in the companion neutronics paper [35].

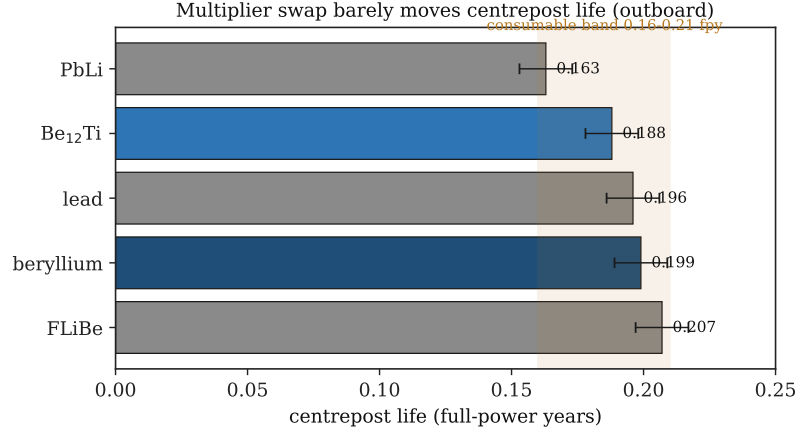


Figure 5. Centrepost life across the material swap, multiplier outboard. All five media overlap within the consumable 0.16–0.21 full-power-year band: the centrepost is shadowed by the inboard shield, which the swap does not change.

The dual-role hypothesis — that a lead multiplier, a better bulk attenuator, could double as inboard shield — is falsified directly by placing the multiplier inboard (Fig. 6): doing so *shortens* centrepost life for every material (beryllium –38 %, lead –46 %, PbLi –25 %), because a breeder-plus-multiplier mantle is a worse fast-neutron shield than the tungsten-carbide stack it displaces, and the lead-bearing materials generate fast neutrons inboard where the conductor cannot afford them. The two functions are correctly kept in separate components: breeding outboard, shielding inboard.

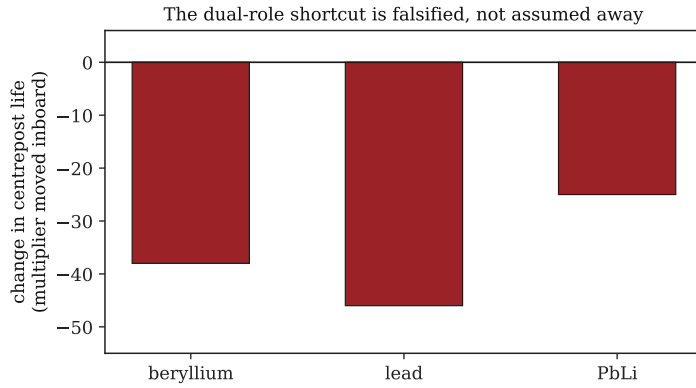


Figure 6. Falsifying the dual-role shortcut. Moving the multiplier inboard shortens centrepost life for every material, because a breeder-plus-multiplier mantle is a worse fast-neutron shield than the tungsten-carbide stack it displaces.

5.4 Activation is prompt-decay dominated and stays below Class C

Solving the Bateman system Eq. (12) on the converged flux (2 full-power years, cooling grid 1 s / 1 d / 1 y / 100 y) shows the beryllium multiplier is activation-benign (Fig. 7). The breeder-zone specific activity falls from 1.83×10^{14} Bq kg⁻¹ at shutdown to 1.12×10^{14} within one second — a 39 % drop driven by the prompt ⁸Be and ⁶He decay that Eq. (12) predicts for the bed — then to 7.69×10^{13} at one year and 2.76×10^{11} Bq kg⁻¹ at one hundred years, by which point essentially all remaining activity is bred tritium held in the ceramic. The associated decay heat falls from 17.9 to 14.5 W kg⁻¹ in the first second, to 0.373 W kg⁻¹ at one day, 0.195 at one year, and

$2.5 \times 10^{-4} \text{ W kg}^{-1}$ at one hundred years. Every blanket component stays at or below NRC Class C with no greater-than-Class-C waste: the largest Class-C index anywhere is the ceramic breeder at 9.4×10^{-3} (driven by ^{14}C), two orders of magnitude below the boundary, and the structural Eurofer sits at Class-A index 0.47 [34]. Tritium management, not structural activation, sets every handling class.

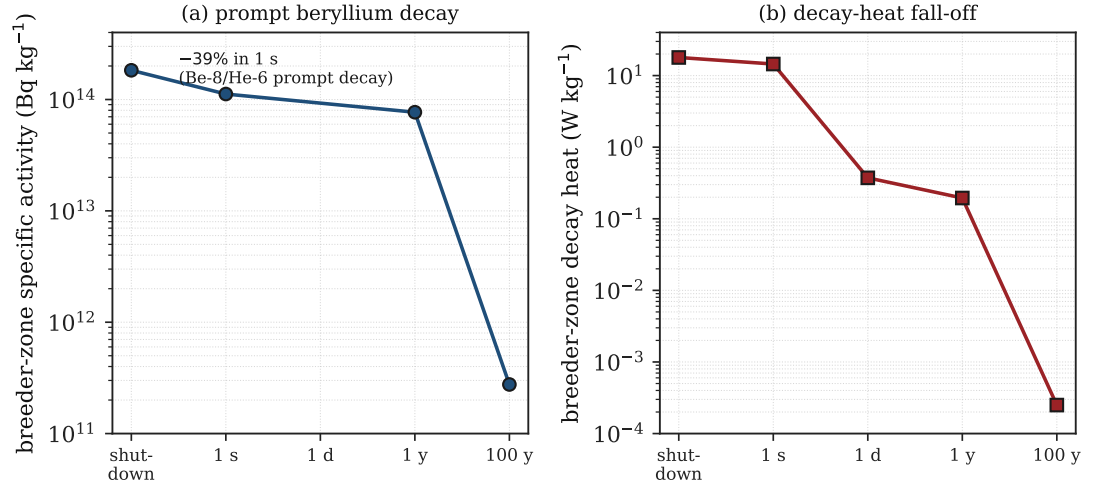


Figure 7. Breeder-zone activation after two full-power years. (a) Specific activity collapses 39 % in the first second from prompt $^8\text{Be}/^6\text{He}$ decay in the beryllium bed, then decays to $2.76 \times 10^{11} \text{ Bq kg}^{-1}$ at 100 y. (b) Decay heat falls more than four orders of magnitude over the same interval.

5.5 The beryllium multiplier lifts the breeding ladder

Placed in its blanket, the beryllium multiplier is what carries the TBR from deficit to closure (Fig. 8). The as-drawn blanket breeds a net 0.742 — below unity, a tritium consumer. The frozen beryllium recipe at 80 % packing lifts that to 1.062; adding a dedicated 5 cm solid-beryllium layer reaches 1.18–1.20; and the canonical breeding anchor at coverage fraction $f_{\text{cov}} = 0.90$ settles at 1.189 — clearing the 1.05 self-sufficiency gate of Eq. (6) with real margin, and held below the 1.42 design ceiling (an even-higher-multiplication optimum of 1.475 was deliberately refused as beyond the ceiling). The admissible window in coverage is mapped in Fig. 9: net TBR runs $1.123 \rightarrow 1.189 \rightarrow 1.215$ across $f_{\text{cov}} = 0.85 \rightarrow 0.90 \rightarrow 0.92$, every point above the gate and below the ceiling.

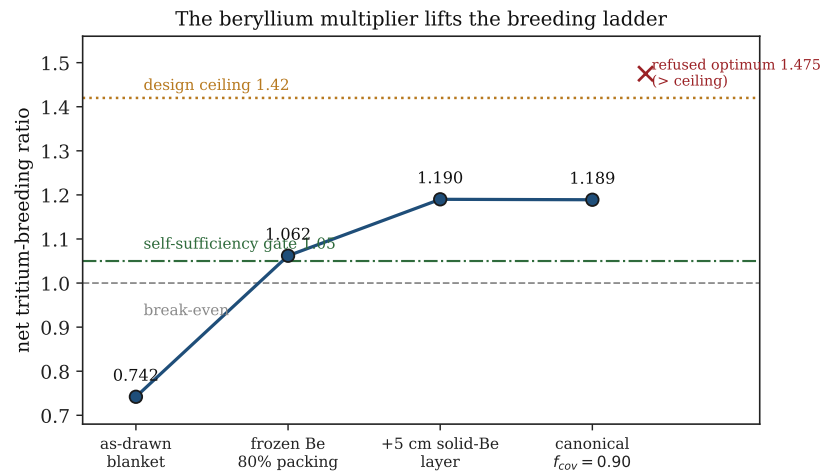


Figure 8. The beryllium multiplier lifts the tritium-breeding ratio from an as-drawn 0.742 deficit to the canonical 1.189, clearing the 1.05 self-sufficiency gate and staying below the 1.42 design ceiling. The 1.475 optimum was refused as beyond the ceiling.

By Eqs. (6)–(7) the margin above the gate becomes a physical surplus (Fig. 10): the self-sufficiency window carries a net surplus of 0.59–1.02 kg-T per full-power year per unit and an

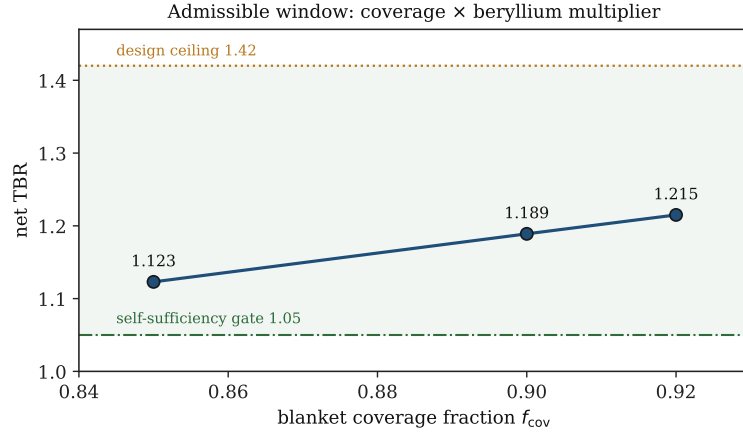


Figure 9. The admissible operating window in coverage fraction. Net TBR rises with f_{cov} and stays between the 1.05 gate and the 1.42 ceiling across 0.85–0.92; the canonical anchor is $f_{cov} = 0.90 \rightarrow 1.189$.

inventory-doubling time of 2.7–4.8 years, while the advanced-blanket ceiling (TBR = 1.42) reaches $\sim 1.99 \text{ kg-T fpy}^{-1}$ per unit, so a two-unit advanced pairing approaches $\sim 4 \text{ kg-T yr}^{-1}$ — the fleet-export basis of the companion analysis [29, 30].

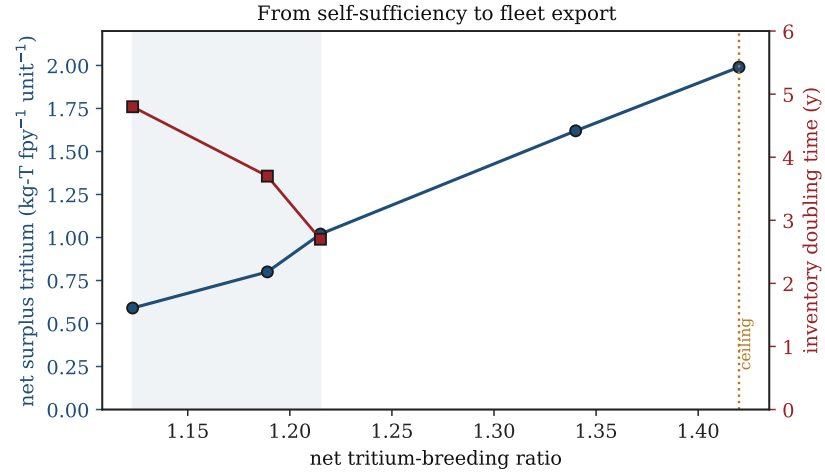


Figure 10. Net surplus tritium (left axis) and inventory-doubling time (right axis) versus net TBR. The self-sufficiency window (1.12–1.22) yields 0.59–1.02 kg-T fpy⁻¹ and doubling in 2.7–4.8 y; the 1.42 ceiling reaches $\sim 1.99 \text{ kg-T fpy}^{-1}$ per unit.

6 Discussion

6.1 Against the published benchmarks

The qualification lands where the fusion-blanket literature would predict but sharpens it for the ST. Beryllium and its beryllides are the reference multipliers of the European HCPB line [11, 16, 17], and the first-principles material review of Hernández and Pereslavytsev [10] ranks Be and Be₁₂Ti ahead of lead precisely on the $(n, 2n)$ -threshold grounds formalised in Eqs. (3)–(4); our 1.054-versus-0.806 closure gap is the quantitative expression of that ranking at the Hyperion operating point. The self-cooled liquids we rule out on breeding grounds (PbLi 0.828, FLiBe 0.908) are the same media that neutronic comparisons for ARC-class high-field devices find competitive only in thicker outboard builds [14, 15]; the ST’s thin, centrepole-constrained radial budget is exactly the regime in which they lose, consistent with our result rather than in tension with it. On the fuel-cycle side, the self-sufficiency inequality Eq. (6) and the ≥ 1.05 engineering gate follow the analyses of Abdou and Sawan [19, 20, 21]: the required TBR margin above unity is set by the loss fraction ε and the start-up inventory, and our canonical 1.189 leaves the headroom those analyses call for. The activation picture — prompt $^8\text{Be}/^6\text{He}$ collapse and a long-time inventory set by ^{14}C and tritium rather than the multiplier — matches the gas-production and transmutation accounting of Gilbert *et al.* [25] and the FISPACT-II inventory framework [6].

6.2 *Why the beryllide is the honest fallback*

The beryllide Be_{12}Ti is not a rejected option but the recorded fallback, and the governing equations say why. Its razor-thin 1.002 closure is the direct consequence of the dilution term in Eq. (3): replacing one beryllium in twelve with titanium lowers N_{Be} and therefore the $(n, 2n)$ yield. What it buys is the swelling robustness of Eq. (11) — a higher operating-temperature ceiling and an intermetallic that resists the bubble coarsening that plants of elemental beryllium accumulate at $\sim 10^4$ appm fpy⁻¹ near the first wall [12, 13, 25]. Choosing the beryllide therefore means recovering the spent margin elsewhere, and the ladder shows exactly where: the coverage lever (Fig. 9) and the ^6Li -enrichment lever both have headroom above the canonical point. The design records beryllium as the qualified multiplier and Be_{12}Ti as the helium-robust fallback whose margin cost is quantified and whose recovery path is mapped — a decision made on the physics, not deferred.

6.3 *Two magnets, one blanket*

The centrepost result reinforces a structural choice that runs through the Kronos design: the breeder centrepost (peak field 16.84 T on the conductor) and the tandem-mirror burner plug (26.49 T) are two distinct magnets with distinct duties, and here the analogous separation is between the multiplier’s breeding duty and the inboard shield’s protection duty. Section 5.3 shows that collapsing those two duties into one component — a lead multiplier doing double service as shield — is not merely suboptimal but actively harmful, shortening centrepost life by up to 46 %. The blanket keeps breeding and shielding in separate radial zones for the same reason the machine keeps its two magnets separate: each function is optimised against a different objective, and folding them together degrades both.

7 Limitations

The reported ratios are beginning-of-life values. The ^6Li burnup that slowly erodes the breeding channel over the operating life, and the helium-swelling kinetics of Eq. (11) that the multiplier accumulates, are stated here as physics and estimated to first order but are not time-integrated in the transport: a coupled depletion study that evolves both is the one named next step. The coverage and enrichment headroom visible in the breeding ladder (Figs. 8–9) is sized precisely to absorb the end-of-life erosion that study will quantify, but until it is run the 1.189 anchor should be read as a beginning-of-life value with a mapped, not yet integrated, margin.

8 Conclusion

A controlled material swap at the closing blanket configuration, solving the linear Boltzmann transport equation in coupled OpenMC on the NVIDIA GPU HPC campaign, qualifies beryllium as the neutron multiplier for the Hyperion breeder, closing tritium at net TBR = 1.054 ± 0.001 . The beryllide Be_{12}Ti is the only credible alternative, holding self-sufficiency at a razor-thin 1.002 and standing as the helium-robust fallback; lead, PbLi, and FLiBe all fall below closure. The ranking is robust to the evaluated-data library at the 0.5–0.6 % level, the multiplier is confirmed a breeding lever rather than a centrepost-life lever — the dual-role shortcut is falsified, not assumed away — and the beryllium multiplier is activation-benign, shedding 39 % of its breeder-zone activity in the first second and staying at or below NRC Class C throughout. In its blanket the multiplier lifts the breeding ratio from an as-drawn 0.742 to a canonical 1.189, clearing the 1.05 self-sufficiency gate with a mapped surplus of 0.59–1.02 kg-T per full-power year and an inventory-doubling time of 2.7–4.8 years, establishing tritium self-sufficiency with margin.

Acknowledgments

Part of the Kronos Fusion Energy 2026 design series. The authors thank the Kronos de-risking team for the independent re-run of the neutron-multiplier lever study.

Reproducibility. The multiplier-ranking, nuclear-data cross-check, centrepost-life, and activation results are frozen anchors (neutron-multiplier lever study; register serials BR-L2-A7, BR-L2-A7u and BV-05) in the Kronos Physics De-Risking Register and are regenerable from the coupled-OpenMC input decks archived with this paper; transport and activation ran on the NVIDIA GPU HPC campaign under deterministic seed 20260726. This paper is archived at doi:10.5281/zenodo.22645889; official version: <https://www.kronosfusionenergy.com/publications/beryllium-and-be-ti-neutron-multipliers-for-a-compact-s.html>; the register is archived at doi:10.5281/zenodo.22133057.

Data availability

This paper is archived at doi:10.5281/zenodo.22645889. The OpenMC models, multiplier-swap and coverage/enrichment lever scripts, activation input decks, and tally outputs are archived with the deposit and reference the Kronos 2026 series and its de-risking register doi:10.5281/zenodo.22133057. Any economic analysis is reported separately and is not included in the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed. The Be₁₂Ti beryllide multiplier route is prior-art-bounded and is recorded as such.

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